جامعة حفر الباطن
University of Hafr Al Batin

Lecture 12 - Domain Name System (DNS)

Objectives

- DNS Overview.
- Need For DNS.
- Name Space.
- Domain Name Space.
- Zones vs. Domains.
- DNS In The Internet.
- DNS Security.

DNS Overview

- DNS is a client/server application program used to help other application programs.
- DNS is used to map a host name in the application layer to an IP address in the network layer

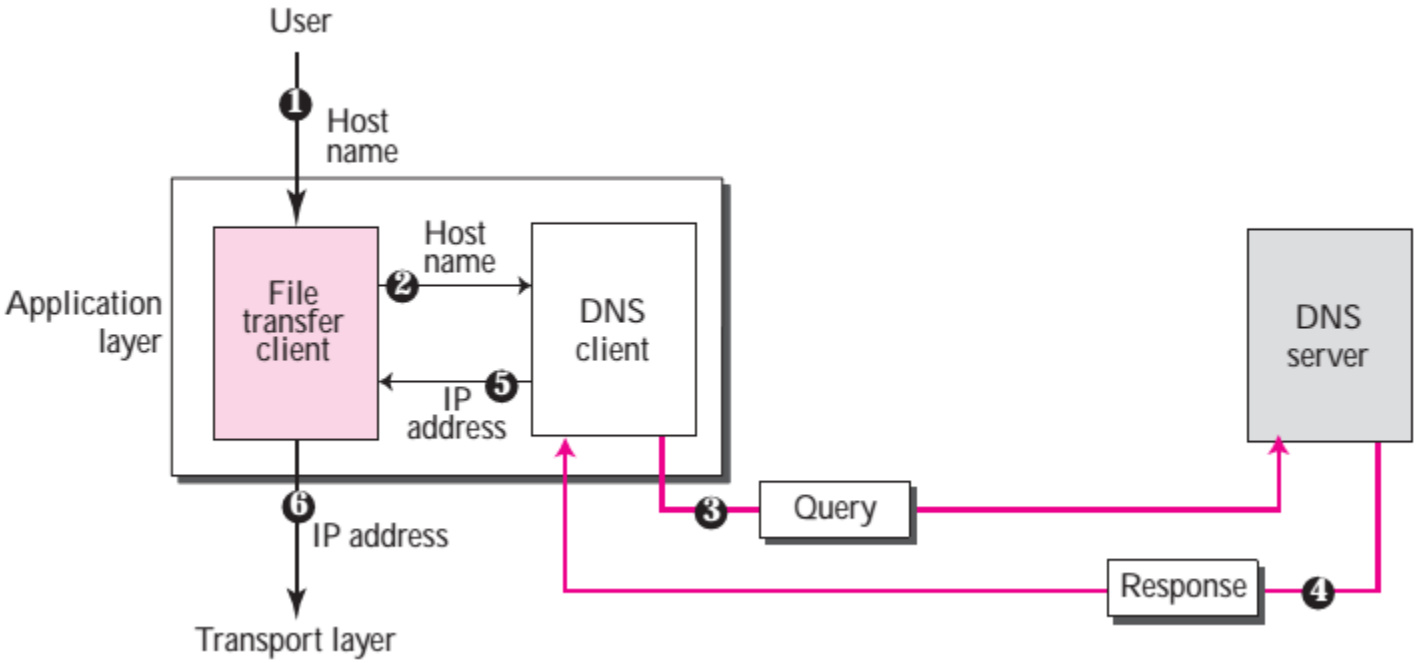
Need For DNS

- To identify an entity, TCP/IP protocols use the IP address, which uniquely identifies the connection of a host to the Internet.
- However, people prefer to use names instead of numeric addresses.
- **Therefore, we need a system that can map a name to an address or an address to a name.**

Need For DNS

- One solution would be to store the entire host file in a single computer and allow access to this centralized information to every computer that needs mapping.
  - But we know that this would create a huge amount of traffic on the Internet.
- Another solution is to divide this huge amount of information into smaller parts and store each part on a different computer.
  - The host that needs mapping can contact the closest computer holding the needed information.
  - This method is used by the Domain Name System (DNS)

Purpose of DNS



Name Space

- To be unambiguous, the names assigned to machines must be carefully selected from a name space with complete control over the binding between the names and IP addresses.
- In other words, the names must be unique because the addresses are unique.
- A **name space** that maps each address to a unique name can be organized in two ways: flat or hierarchical.

Name Space

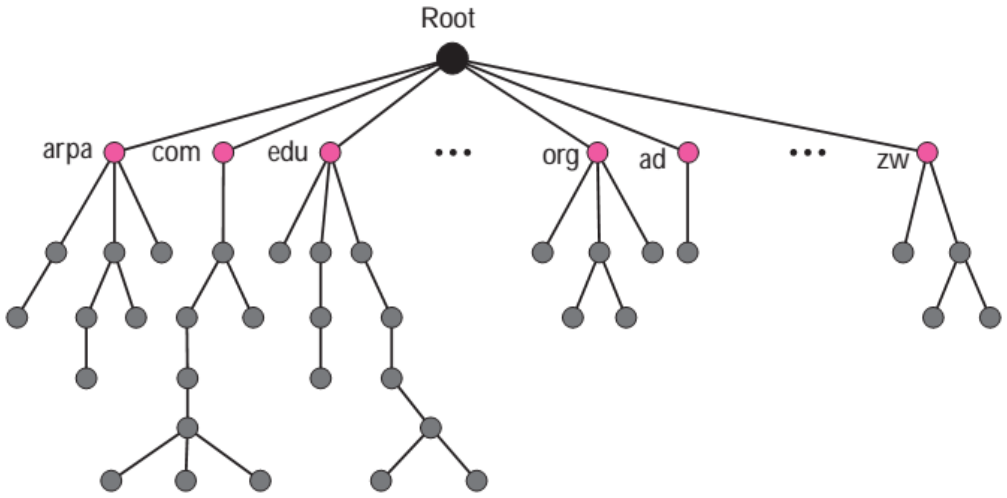
- Flat Name Space:
  - A name is assigned to an address.
  - A name in this space is a sequence of characters without structure.
  - The names may or may not have a common section; if they do, it has no meaning.
  - The main disadvantage of a flat name space is that it cannot be used in a large system such as the Internet
    - because it must be centrally controlled to avoid ambiguity and duplication.

Name Space

- Hierarchical Name Space:
  - Each name is made of several parts.
    - The first part can define the nature of the organization.
    - The second part can define the name of an organization.
    - The third part can define departments in the organization.
    - And so on…
  - The authority to assign and control the name spaces can be *decentralized*.
    - A central authority can assign the part of the name that defines the nature and the name of the organization.
    - The responsibility of the rest of the name can be given to the organization itself.
    - The central authority controls only part of the name, not the whole.

Domain Name Space

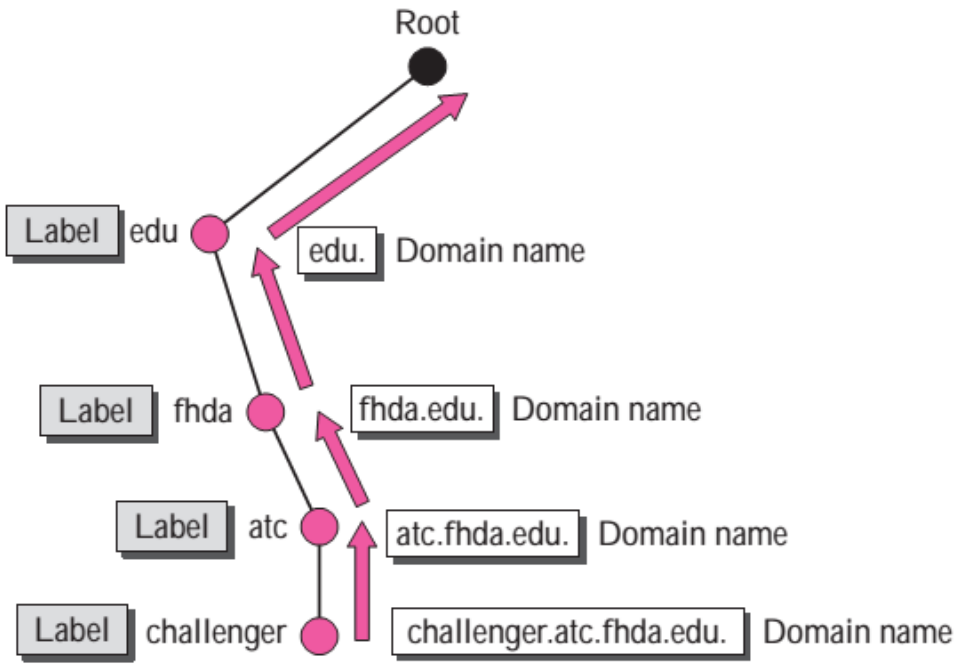
- To have a hierarchical name space, a domain name space was designed.
- In this design, the names are defined in an inverted-tree structure with the root at the top.
- The tree can have only 128 levels: level 0 (root) to level 127.



Domain names and labels

- Each node in the tree has a **label**.
  - Maximum of 63 characters.
  - Root label is a null string (empty string).
  - Children of a node <u>must</u> have different labels.

- Each node in the tree has a **domain name**.
  - A sequence of labels separated by dots (.).
  - Domain names are always read upward.



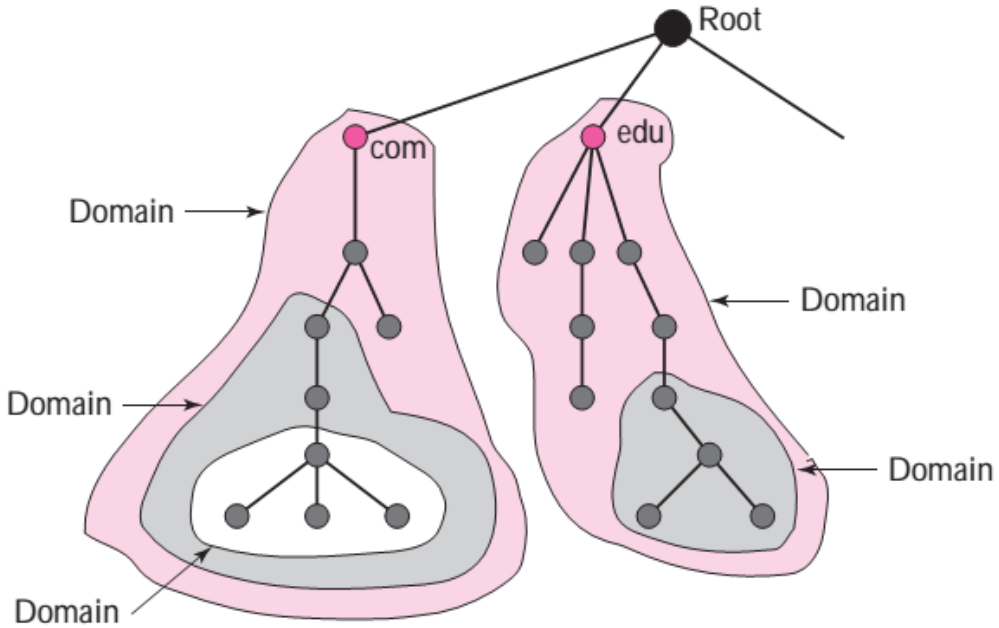
Domain names and labels

- **Fully Qualified Domain Name (FQDN):** is a domain name that contains the full name of a host.
  - It contains all labels, from the most specific to the most general.
  - The name must end with a null label, but because null means nothing, the label ends with a dot (.).
- **Partially Qualified Domain Name (PQDN):** is a domain with a label that is not terminated by a null string.
  - A PQDN starts from a node, but it does not reach the root.
  - used when the name to be resolved belongs to the same site as the client

FQDN	PQDN
challenger.atc.fhda.edu. cs.hmme.com. www.funny.int.	challenger.atc.fhda.edu cs.hmme www

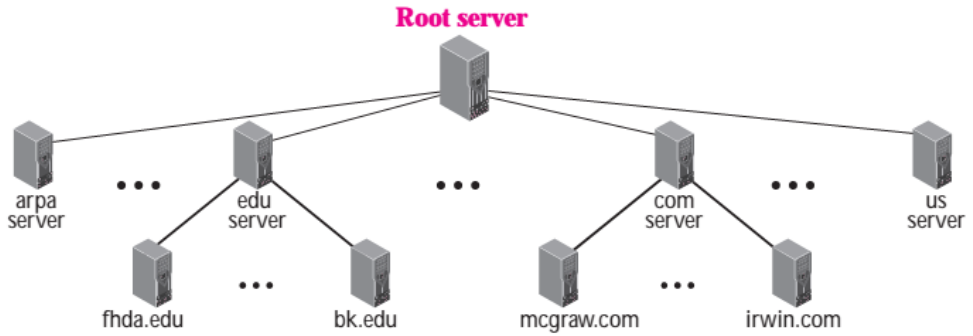
Domain

- **A domain is a subtree of the domain name space.**
- The name of the domain is the name of the node at the top of the subtree.



Distribution of Name Space

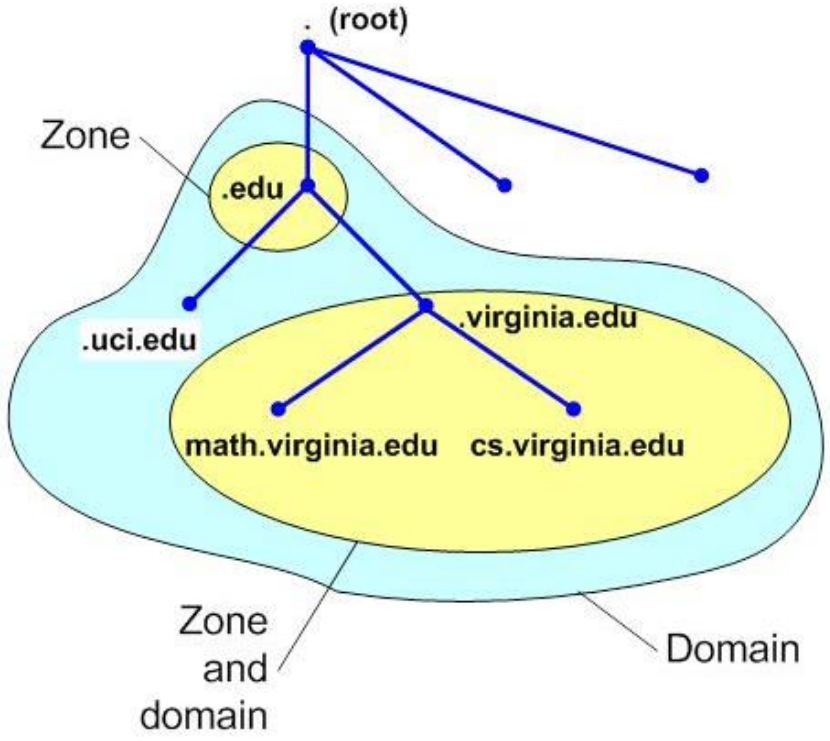
- The information contained in the domain name space must be stored.
- However, it is very inefficient and also not reliable to have just one computer store such a huge amount of information. **Why?**
  - Responding to requests from all over the world places a heavy load on the system.
  - Any failure makes the data inaccessible.
- **The solution:** distribute the information among many computers called DNS servers.



Zones vs. Domains

- **Zone:** is what a server is responsible for or has authority over.

- If a server does not divide the domain into smaller domains, the “domain” and the “zone” refer to the same thing.
- The server makes a database called a *zone file* and keeps all the information for every node under that domain.
  - It may delegate the information to lower-level servers.



Root Server

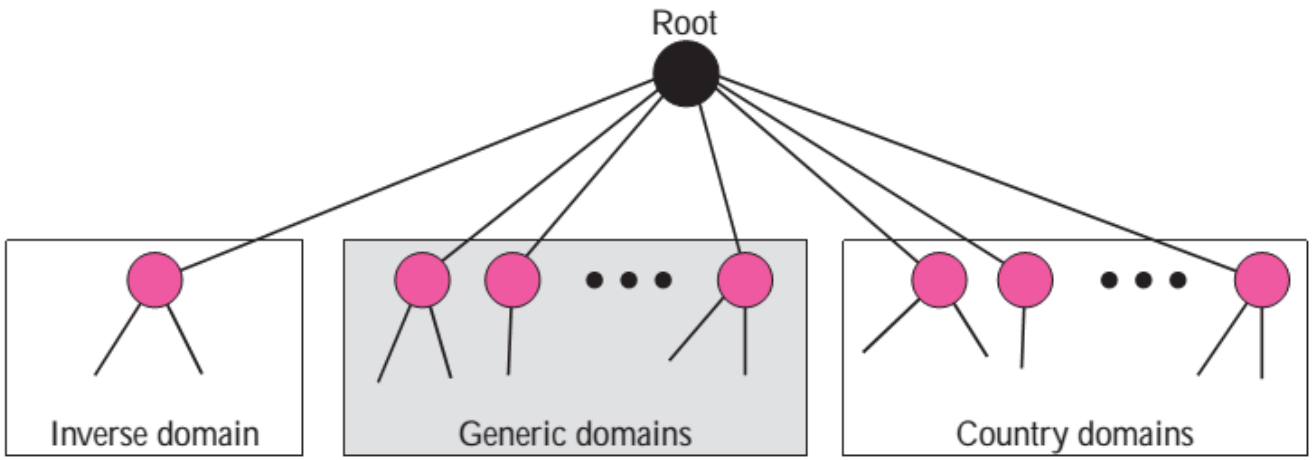
- **A Root Server: is a server whose zone consists of the whole tree.**
- A root server usually does not store any information about domains but delegates its authority to other servers, keeping references to those servers.
- There are several root servers, each covering the whole domain name space.
- Root servers are distributed all around the world.

Primary and Secondary Servers

- **Primary Server:** is a server that stores a file about the zone for which it is an authority.
  - It is responsible for creating, maintaining, and updating the zone file.
  - Loads all information from the local disk.
- **Secondary Server:** is a server that transfers the complete information about a zone from another server (primary or secondary).
  - Loads all information from the primary server.
  - The secondary server neither creates nor updates the zone files.
  - When the secondary downloads information from the primary, it is called zone transfer.
- **The idea is to create redundancy for the data.**
  - So that if one server fails, the other can continue serving clients.

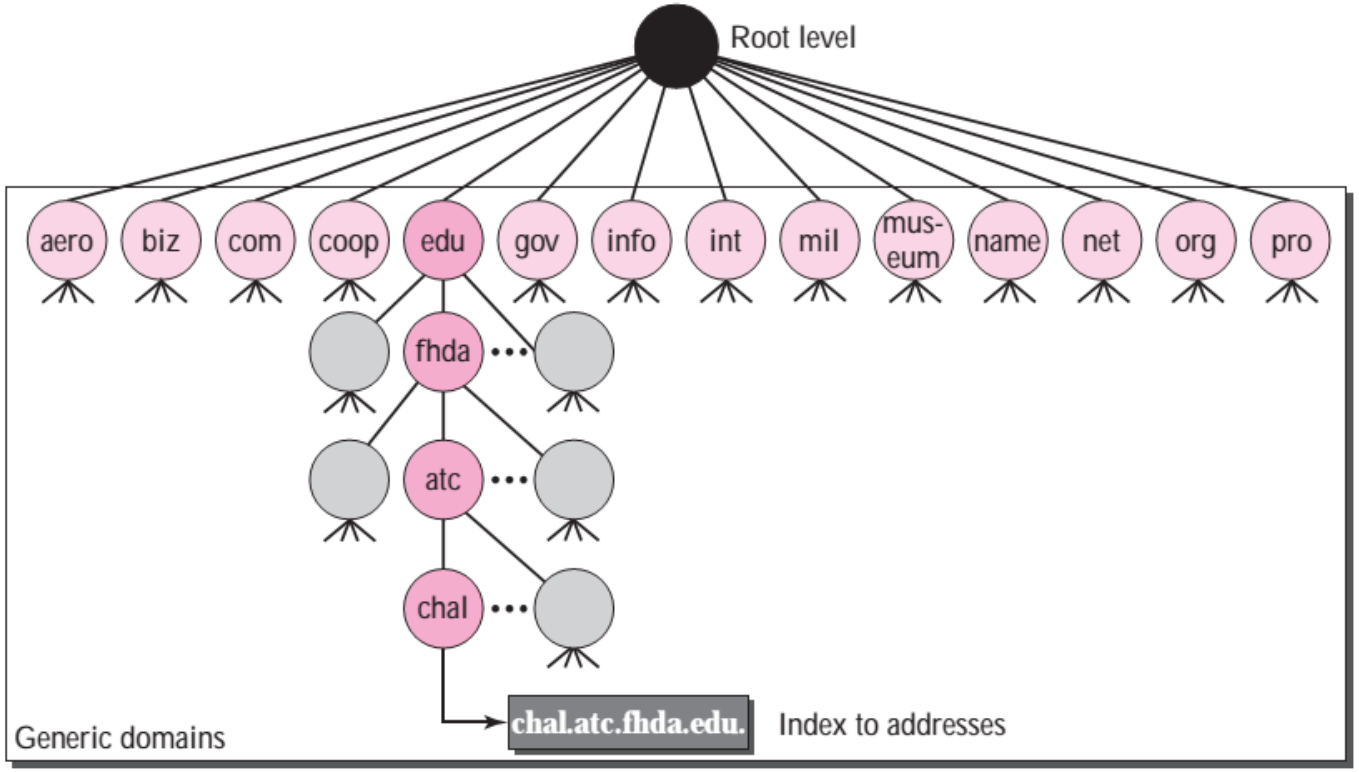
DNS In The Internet

- DNS is a protocol that can be used in different platforms.
- In the Internet, the domain name space (tree) is divided into three different sections:
  1. Generic domains.
  2. Country domains.
  3. Inverse domain.



Generic Domains

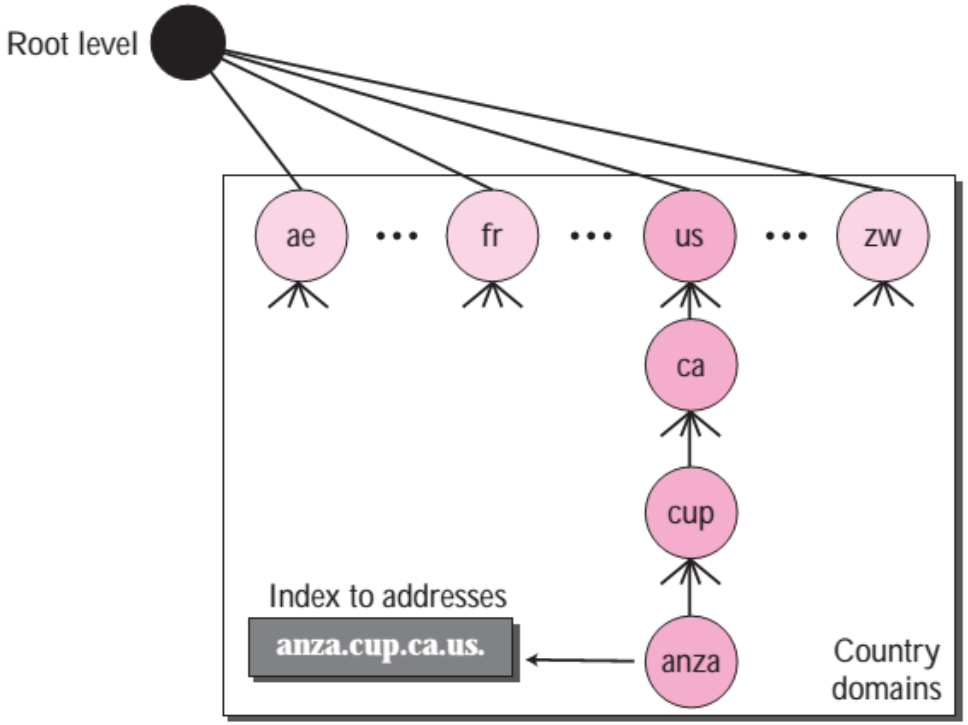
- The **generic domains** define registered hosts according to their generic behavior.



Label	*Description*
aero	Airlines and aerospace companies
biz	Businesses or firms (similar to "com")
com	Commercial organizations
coop	Cooperative business organizations
edu	Educational institutions
gov	Government institutions
info	Information service providers
int	International organizations
mil	Military groups
museum	Museums and other non-profit organizations
name	Personal names (individuals)
net	Network support centers
org	Nonprofit organizations
pro	Professional individual organizations

Country Domains

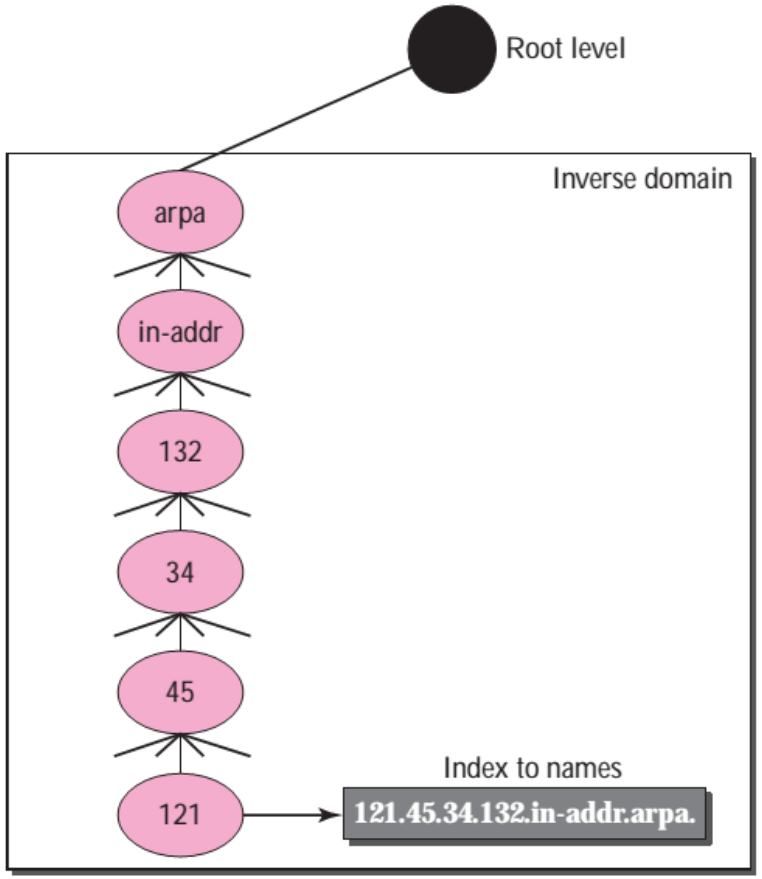
- The **country domains** section uses two-character country abbreviations.



De Anza College in Cupertino in California in the United States

Inverse Domain

- The **inverse domain** is used to map an address to a name.
  - The inverse domain is added to the domain name space with the first-level node called arpa (for historical reasons).
  - The second level is also one single node named in-addr (for inverse address).
  - The rest of the domain defines IP addresses.



Registrar

- **How are the new domains added to DNS?**
  - This is done through a registrar.

- **A Registrar: is a commercial entity accredited by ICANN.**
- A registrar first verifies that the requested domain name is unique and then enters it into the DNS database.
  - Usually, a fee is charged.

DNS Resolution

- **Name-address Resolution:** is the mapping of a name to an address or an address to a name.
- a DNS client called a **resolver**.
  - The resolver accesses the closest DNS server with a mapping request.
- **Mapping Names to Addresses:**
  - The resolver gives a domain name to the server and asks for the address.
  - The server checks the generic domains or the country domains.
- **Mapping Addresses to Names:**
  - The resolver can send an IP address to a server to be mapped to a domain name.
  - The server checks the inverse domain.

DNS Security

- DNS is one of the most important systems in the Internet infrastructure.
  - For example: Web access and e-mail are heavily dependent on DNS.
- DNS can be attacked in several ways including:
  1. The attacker may flood the DNS server to overwhelm it or eventually crash it.
  2. The attacker may read the response of a DNS server.
  3. The attacker may change the response of a DNS server.
- **To protect DNS, IETF has devised a technology named DNS Security (DNSSEC):**
  - It provides the message origin authentication and message integrity using a security service called **digital signature**.